Game Design Document

Journia

Naz Camiş 20211314017

1. Introduction

Language learning can often feel like a daunting task, but what if it became an adventure? Journia is an innovative, story-driven educational game designed to make learning a new language not just effective but also enjoyable. Combining interactive gameplay, rich storytelling, and practical language exercises, Journia provides players with an immersive environment where language acquisition feels natural and rewarding.

2. Story

The game follows the life of a child born into a Turkish family that has recently moved to England. At home, the character grows up speaking Turkish, surrounded by their family's culture and traditions. However, the outside world introduces them to English, a language that is new and essential for adapting to their environment.

As the character grows, the central focus of the game becomes their journey of learning English. From recognizing simple words as a toddler to mastering complex vocabulary and grammar, the player experiences the challenges, rewards, and milestones of becoming bilingual. The game intertwines this language-learning process with the character's growth, showing how English gradually becomes a bridge to new friendships, opportunities, and self-expression in a foreign land.

3. Our Character and Development

3.1 Customization

Starting the Game

Players begin the game as a baby. At this stage, personalization is minimal, allowing gender and skin tone selection of the character.

Customization Progression

As the character progresses through life stages, additional customization options become available. These changes reflect the character’s growth and the player’s increasing control over their avatar:

Physical Features: Hair color, hairstyle, eye color and body type.

Clothing and Accessories: Shirts, pants, shoes, hats and other accessories.

Dynamic Customization Through In-Game Activities

Customization isn’t limited to static menus; it’s tied into the game world to enhance immersion.

Hair Styling:

Players can visit barber shops or salons to change hairstyles or hair colors. Each option may have an in-game currency cost, making it a rewarding achievement for players.

Shopping for Clothes:

In certain life stages, players can visit clothing stores to purchase new outfits.

Stores offer styles appropriate to the character's life stage, such as casual wear for teens or business attire for adults.

Unlockable Styles and Items:

Some items or styles could be unlocked by reaching specific milestones, leveling up, or forming relationships with NPCs receiving gifts or rewards.

3.2 How Life Stages Affect Our Character

The life stages in the game not only represent a natural progression of growth but also unlock new environments and gameplay opportunities. Each stage is tied to a specific duration, ensuring players immerse themselves in unique experiences and challenges.

Daily Time Progression: Each day lasts a limited duration, allowing interactions with family members and household objects.

•Baby

Stage Duration: 5 in-game days.

•Childhood

New Opportunities: Players can attend elementary school, interact with classmates, and explore new environments.

Stage Duration: 10 in-game days.

•Teenage Years

New Opportunities: The player builds friendships, attends events, and navigates complex NPC interactions.

Stage Duration: 10 in-game days.

•Adulthood

New Opportunities: The player earns in-game currency, builds relationships, and learns advanced vocabulary for professional and social contexts.

Stage Duration: 20 in-game days.

•Retirement

New Opportunities: Reflective gameplay allows revisiting key memories and teaching others.

Stage Duration: 10 in-game days.

3.3 How Our Character Levels Up

The leveling-up mechanism in our game is designed to reward players for actively learning and engaging with the game environment. Here’s how the character progresses through levels:

3.3.1 Gaining XP (Experience Points):

Players earn XP through various activities, such as:

•Learning New Words: Each correctly learned word or phrase grants XP. More complex vocabulary awards higher XP.

•Completing Daily Quizzes: At the end of each in-game day, the player answers questions based on the words they learned that day. Quiz performance determines the amount of bonus XP awarded.

•Interacting with Objects and NPCs: Engaging with objects or NPCs that introduce new vocabulary or scenarios earns XP.

•Mini-Tasks and Challenges: Completing small in-game tasks or challenges provides additional XP.

3.3.2 Level Progression:

Each age stage has an XP threshold. For example, to move from babyhood to childhood, players must reach a specific XP target while also completing the required in-game days for that stage.

If the XP target is not met by the end of the stage's time limit, extra days are granted for players to reach their goal.

3.3.3 Daily XP Bonus:

Simply completing a full in-game day rewards players with a set amount of XP, encouraging consistent progress.

3.3.4 Age-Based Challenges:

As the character grows, leveling up becomes more challenging. Players are required to learn more advanced vocabulary, complete more complex interactions, and perform better in quizzes to earn XP.

3.4 Daily Time and Gameplay Flow

Day Progression: Each day is limited in time, providing players with a structured experience.

Stage Transitions: At the end of each life stage, players complete a summary task to unlock the next phase of the game.

By tying new maps and opportunities to the life stages, the game ensures a sense of discovery and progression, keeping players engaged while seamlessly integrating educational elements

3.5 Character Movement and Controls

Movement Controls:

W, A, S, D Keys: Used to move the character forward, backward, left, and right respectively, allowing full navigation of the game world.

Mouse Interaction:

Mouse: Used for interacting with objects, NPCs, and the environment, enabling players to learn vocabulary, initiate conversations, and perform actions.

Menu and Progress Tracking:

Esc Key: Opens the options menu, allowing the player to resume the game, access settings to adjust audio, graphics, and controls, or quit the game to return to the desktop or main menu.

M Key: Opens the Map to show key locations in the current environment, helping players navigate the game world.

T Key: Opens the Phone, which features a Relationship Control Panel. Players can view and manage their relationships with various NPCs. Receive messages or notifications from in-game characters.

C Key: Opens the Character Screen, where players can view and track their character's progress, appearance, achievements, and other key information (level, XP, stats, etc.).

J Key: Opens the language journal, where players can track learned vocabulary, check progress, and see completed tasks.

Movement Enhancements:

Shift Key: Enables the character to run faster.

3.6 Character Interaction with NPCs

In the game, the character can interact with NPCs (Non-Playable Characters) in various ways, depending on the stage of their life and the environment they are in. The interaction process works as follows:

Dialogue System: When the character approaches an NPC, a dialogue box will appear on the screen. Players can choose from multiple dialogue options to respond or continue the conversation. These options will vary

depending on the situation and the player's progress in the game.

Learning New Words and Phrases: NPCs will provide new vocabulary by speaking to the character. These words will be highlighted or shown in English, and players will learn them as they interact. For example, when the character speaks with their parent or teacher, they will encounter common phrases related to daily life.

Quizzes and Tasks: In some interactions, NPCs may ask the character questions or assign simple tasks (e.g., "What is your name?" or "Can you find the cat?"). Completing these tasks will earn XP points and help the player progress in the game.

Emotion and Relationship Building: The way the player responds to NPCs may also influence the relationship between the character and the NPC. Positive interactions will strengthen relationships, while negative responses may impact NPC attitudes toward the character.

These interactions will be essential for the player to improve their English skills while also advancing the character's personal growth and progress through the game.

4. Game Universe

4.1 Maps

4.1.1 Baby Stage (Infancy):

Room: A single room map where the player interacts only with their parents

4.1.2. Childhood Stage:

House Map: Includes living room, bathroom, kitchen, parents bedroom.

Street Map: Used to travel to school. Features a convenience store across the street. Includes NPCs like a child playing outside and an elderly woman (changes daily).

School Map:

•**Hallway:** Principal, other teachers, or students may appear on different days.

•**Classroom:** Includes classmates and a teacher NPC.

Park Map: A public park with other kids and parents NPCs.

4.1.3 Teenage Stage (High School):

House Map: Same as in childhood, but the bedroom is upgraded to a teenager’s room. Includes a new sibling (baby NPC).

Street Map: Same layout, but the convenience store owner is replaced by their son (previous owner is implied to be gone).

High School Map:

Hallway: Rotating appearances of the principal, students, and other teachers.

Classroom: Includes old childhood classmates (grown up) and a new teacher.

Club Room: A theater club space with a teacher and rotating NPCs for activities.

Park Map: Remains unchanged.

Arcade Map: A game center with small mini-games like Tetris.

4.1.4 University Stage:

Dorm Room Map: Shared with one random roommate NPC.

Campus Map: Includes: University garden, library, cafeteria, **l**ecture hall.

4.1.5 Adulthood Stage:

Apartment Map: If married, shared apartment with partner; If single solo apartment.

Workplace Map: Office or work environment depending on the chosen career.

Restaurant Map: A place for dining and social interactions.

Hairdresser and Clothing Store Maps: Players can customize their appearance and outfits.

Childcare Option: Midway through adulthood, the option to have a child is unlocked, introducing interactions with the child NPC.

4.1.6 Elderly Stage:

Country House Map: A quiet rural home where the player lives.

Garden Map: A serene outdoor space for gardening and relaxing.

Farm Map: Features a small countryside farm with animals.

Local Market Map: A nearby market for daily necessities.

4.2 Time:

Day/Night Cycle:

The game have a dynamic day and night system, with a set period of time dedicated to each cycle. Day last 10 minutes and night last 5 minutes. The time of day will affect gameplay, NPC behavior, and available activities. Some events, quests, and NPCs will only be available during the day or night.

S**easonal Changes:**

The game world will experience changing seasons, with each season bringing different weather patterns and environmental changes. Each season's unique theme, event and festival will also provide extra chances to socialise. For instance:

•**Spring:** Flowers bloom, and the weather is mild. It will be a spring festival.

•**Summer:** Longer days and warmer weather, with people spending more time outdoors.

•**Autumn:** Leaves fall, and it gets colder.

•**Winter:** Snow falls, and certain areas might be blocked due to cold conditions.

Character Age Progression:

Time also progresses with the character's age, affecting their appearance, abilities, and interactions. This is synced with life stages, where each stage of life progresses over a set amount of time in the game.

Age-Related Changes: As time passes, the character will experience physical changes, such as growing taller or older, and gaining new skills or experiences.

Real-Time Events:

•Some in-game events or quests can be based on the actual in-game time. For example, certain activities or NPCs may only be available at certain times of the day or on specific days of the week.

•There is scheduled events (like festivals, public holidays, or NPC birthdays) that the player can look forward to, creating a sense of time-bound progression.

Time Management:

The player will have to manage their character's time wisely. They can schedule activities, attend events, or spend time interacting with NPCs. Time is tracked with a clock and calendar in the game interface, giving the player an overview of their day-to-day progression.

Time-Limited Goals/Quests:

Some quests or objectives have a deadline. The player will need to manage their time to accomplish tasks before the deadline passes.

Fast-Forward/Skip Time:

Going to bed at night triggers the end of the current in-game day. Upon sleeping, the player's character wakes up in the morning of the next day.Players can choose to sleep early and skip a time period. Skipping time by going to sleep may cause the player to miss certain interactions. For example, if the player skips too many days, they might miss important NPC interactions, events, or quests that were time-sensitive. Also leveling up may be harder.

NPC Schedules:

NPCs will follow their own schedules based on time. The player will need to track these schedules to interact with specific NPCs at the right times.

4.3 World Physics

Collisions:

•The world has a collision system, meaning characters cannot move through solid objects like walls, trees, or buildings. If the character tries to walk into an object, they will stop at the boundary and must find another way around.

•The environment is structured, and movement is restricted by physical objects in the world.

Gravity:

•The world is governed by basic gravity, so the character will stay on the ground unless they are interacting with something.

Object Interaction:

•The environment is interactive but simplified. The character can walk up to items, doors, and other objects to trigger specific actions, such as opening doors or using items.

•No complex interactions like picking up objects or using physics-based movement.

Navigating Obstacles:

•As mentioned earlier, when the character encounters an obstacle, they must manually change their direction to continue moving. These obstacles can include buildings, fences, or natural features like rocks or trees.

•Players can change direction to avoid these obstacles but will need to find alternative routes to move forward in certain situations.

Environmental Effects:

•In certain cases, the environment can influence the character's movement. For instance, muddy terrain might slow down the character's speed, or rain might make the ground slippery.

•These effects are designed to create a simple, interactive world that feels alive without complex physics systems.

5. Quiz System

The quiz system in the game serves as an interactive reinforcement mechanism that enhances players' language learning experience while integrating seamlessly into gameplay.

5.1 Gaining Points to Level Up

Players earn points by answering quiz questions correctly. These points are essential for leveling up and unlocking new maps, vocabulary, and gameplay features. Correct answers grant points encouraging accuracy and focus. Daily quiz limits are tied to player interactions; the more the player engages with their environment, the more questions they unlock. This design motivates exploration and active learning.

5.2 Triggering and Displaying Questions

Quiz questions are triggered under the following conditions:

During Gameplay: Whenever players interact with new objects or NPCs and encounter new vocabulary or sentences.

End-of-Day Pop Quiz: At the conclusion of each in-game day, a review quiz is presented, featuring words and phrases learned that day. This pop quiz consolidates the player's knowledge and offers additional points.

Additionally, the system identifies questions that players struggle with. These questions appear more frequently in subsequent quizzes to ensure reinforcement and mastery.

Engaging with Objects

Interaction with the environment directly contributes to the quiz system. For instance, touching an object like a ball in the baby stage introduces the word "ball." Later, this word may appear in quizzes. This approach not only teaches vocabulary in context but also links learning to immersive gameplay. Repeated incorrect answers to certain questions increase their frequency in quizzes to strengthen the player's retention.

6. Relationship System

The relationship system is designed to simulate real-life social interactions, providing players with meaningful engagement with NPCs while reinforcing language learning.

Building Relationships

Players can strengthen or weaken relationships with NPCs based on their responses during interactions. When initiating contact, players select the type of interaction they wish to pursue, such as friendship, romantic interest, or confrontation. These choices determine the dialogue options available and influence the nature of the relationship.

Interaction Mechanics

Selecting Interaction Type: Players are presented with choices such as small talk, invitations, or arguments. The choice determines the dialogue's tone and intent.

Translation Challenges: To proceed with interactions, players must translate Turkish sentences into English. Correct translations improve the relationship score, while incorrect translations may reduce it.

Maintaining Social Needs

The game emphasizes the importance of social connections by integrating a social needs system. NPC relationships directly impact the player's overall satisfaction and progression. Strong relationships unlock unique rewards, dialogue options, and potential quests, making them a valuable part of the gameplay experience.

This system combines language practice with emotional engagement, ensuring that players' social interactions remain dynamic and educational.

7.Graphics, game design

7.1. Color Palette

Each life stage features a color palette designed to match its tone and atmosphere:

•**Infancy:** Pastel colors (pink, light blue, soft green) create a warm and safe environment.

•**Childhood:** Bright colors (yellow, orange, vibrant blue) are used to evoke a sense of curiosity and playfulness.

•**Adolescence:** Modern and slightly more serious tones (gray, dark blue, white) reflect a transition to maturity.

•**Adulthood:** Neutral and sophisticated colors (gray, burgundy, brown) emphasize professionalism.

•**Old Age:** Warm tones (olive green, cream, gold) are chosen to evoke a nostalgic atmosphere.

7.2 Art Style

•The game adopts a cozy and warm art style, emphasizing soft textures, gentle lighting, and inviting environments. Each life stage is designed to feel comforting and relatable, with an aesthetic that draws players into a world that feels both familiar and magical.

Animation

•Smooth and fluid animations for character movements, interactions, and dialogues.

•Unique animations for language-learning moments, such as a glowing effect when new vocabulary is learned.

Environment Design

•Highly interactive environments with objects that provide language-learning opportunities.

•Dynamic elements such as weather changes, time of day, and seasonal transitions to create a living world.

Accessibility Features

•Colorblind-friendly palettes and adjustable contrast settings.

•Adjustable font size and text-to-speech options for better accessibility.

7.3 Sound Design

•Background music tailored to each life stage, from soothing lullabies in infancy to upbeat tracks in adulthood.

•Sound effects for object interactions and feedback during quizzes.

7.4 User Experience (UX)

•A simple and intuitive user interface that guides players through their learning journey without overwhelming them.

•Context-sensitive tips and tutorials to help new players adapt quickly.

7.5 Reward System

•Visual and audio rewards such as celebratory animations or sounds when the player successfully learns new words or completes tasks.

7.6 Camera Perspective

•Flexible camera angles to provide a better view of the environment and objects of interest.

•Zoom-in feature for closer inspection of interactable objects.

8. Platform, Genre and Target Audience

8.1 Platform

•**PC (Steam):** The game is optimized for high-resolution graphics and broad accessibility, making it suitable for a wide range of users.

•**Mobile (iOS and Android):** A mobile-friendly version is designed for users who want to learn languages on the go.

8.2 Genre

•An educational and interactive story-driven game that combines learning with engaging storytelling.

8.3 Target Audience

•**Age Range:** 8-25 and above.

•**Language Proficiency:** Beginner to intermediate English learners.

•**User Profile:** Children, teenagers, and adults who want to make language learning an enjoyable and immersive experience.

9.Why play this game

Combines Education and Entertainment

Language learning is often perceived as a tedious process. However, this game transforms the experience by sparking the player's curiosity and desire to explore, making language learning enjoyable and engaging.

Real-Life Connections

Each stage incorporates words and dialogues that players might encounter in real-life situations. For example:

Cultural Awareness

Players get to experience the life of a Turkish individual living in England, discovering cultural differences and similarities, and gaining a deeper understanding of cross-cultural dynamics.

10.Visual examples

What is this object?

11.Game Ending

After the retirement stage, the character has lived a peaceful and fulfilling life, leaving behind a legacy of knowledge from their lifelong journey of language learning. In the final scene, the character’s grandchildren are exploring an old attic or a dusty bookshelf when they stumble upon a language journal. This journal contains the words, stories, and notes the player collected throughout the game.

One of the grandchildren opens the journal, and a heartfelt conversation begins:

Grandchild 1: "Grandma/Grandpa learned so much! Look at all these words and notes they wrote down."

Grandchild 2: "They worked so hard to learn languages. Maybe we can follow in their footsteps."

Final Message

The screen fades to black, and a message appears:

"Knowledge and love transcend generations. Learning a new language is not just a skill—it’s a legacy. This journey never truly ends."